

T1.4 Building a lateral root through differentiated barriers in *Brachypodium distachyon* – MALAI Marius

Marius Malai¹, Corentin Maslard¹, Joop E. M. Vermeer¹

¹University of Neuchâtel

Lateral root formation requires coordinated developmental changes across several root tissues. Most of what we know comes from *Arabidopsis thaliana*, and comparatively little is understood about how this process works in grasses. In *Brachypodium distachyon*, the endodermis takes on a different role during lateral root emergence: it re-enters the cell cycle and contributes to the organogenesis via dedifferentiation into the lateral root cap. At the same time, the exodermis forms a resistant layer with extensive ligno-suberin cell wall modifications that must be traversed. To address these gaps, we developed genetic tools that allow us to follow how different tissues contribute to lateral root formation in a model grass root system. We used a Cre/lox lineage-tracing approach to fluorescently mark the endodermis prior to lateral root initiation. Because the fluorescent nuclear label is inherited by all descendant cells, we can track these lineages throughout lateral root organogenesis. Pairing this system with single nuclei RNA sequencing should make it possible to capture the transcriptional shifts underlying endodermis dedifferentiation during lateral root development. In parallel, we generated an exodermis-specific marker line to examine how this tissue accommodates emerging lateral roots. As the outer apoplastic barrier between the root and the soil environment, the exodermis represents an additional differentiated layer that must respond during organ emergence. Together, these tools provide a framework for dissecting how multiple root tissues coordinate to build a lateral root in grasses.